Bayesian Linear Regression

A First Really Useful Bayesian Model

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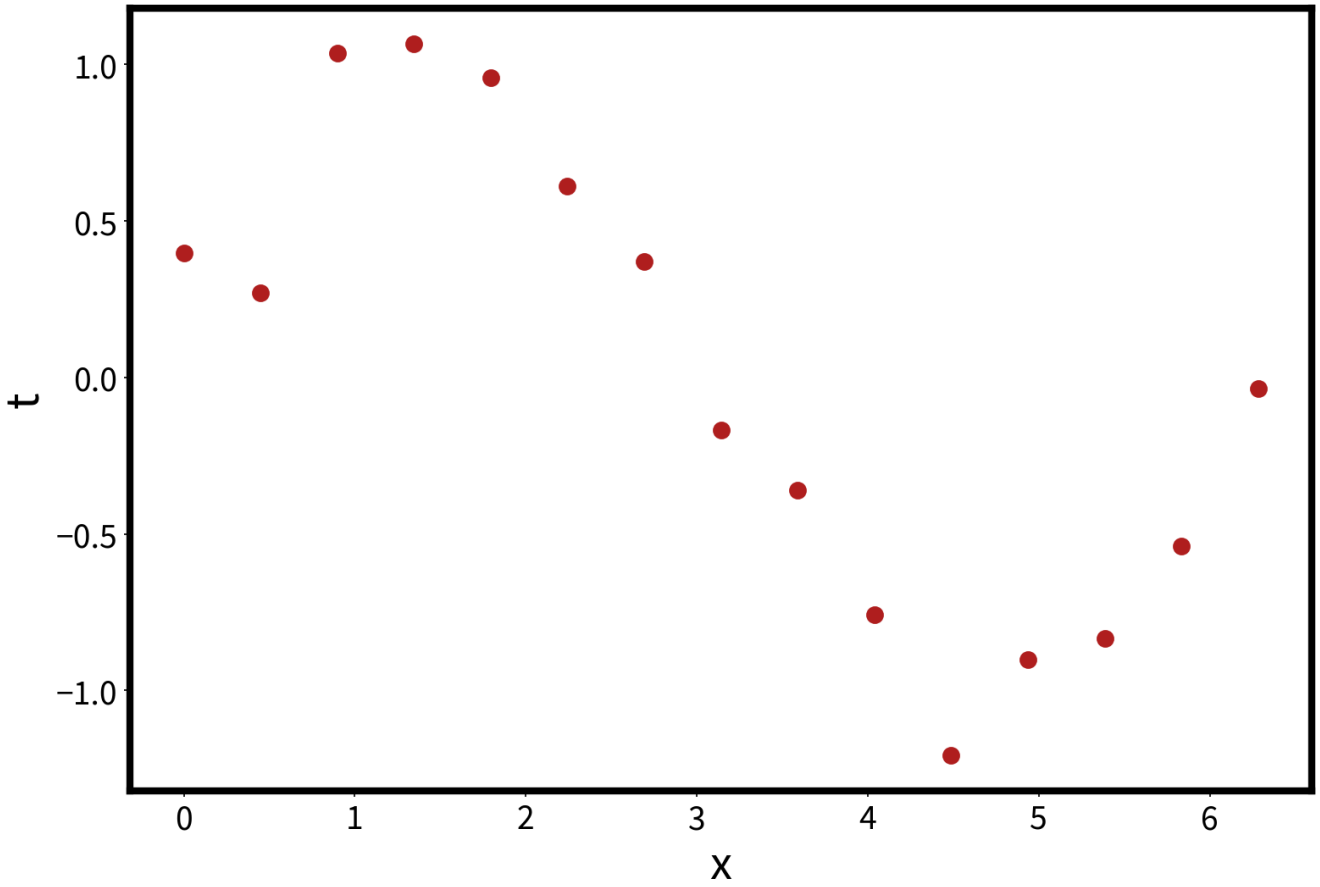
Slide 1 In this Episode ...

- An example ‘regression’ (real-value prediction) problem
- Starting non-Bayesian
  - Linear (in the parameter) models and least-squares
  - Penalised least-squares and regularisation
  - The probabilistic framework
- Going Bayesian
  - Maximum *a posteriori* (MAP) estimation
  - Step-by-step demo of Bayesian “learning”

Example Data Set

Slide 2 Example Regression Problem: Data Set

- Example data set $\mathcal{D}$ of $N = 15$ points generated from the function $f(x) = \sin(x)$ with added Gaussian noise of $\sigma = 0.25$:



- Input variables are denoted x_n and observation ("target") values are denoted y_n
- Ultimate objective: infer the underlying function $f(x)$ from the data set

Linear Models

Slide 3 Linear (in-the-parameters) Regression

- We will model $f(x)$ with a parameterised function $\widehat{f}(x; \mathbf{w})$
- Here, we consider models which are a linearly-weighted sum of M fixed basis functions $\phi_m(x)$:

$$\widehat{f}(x; \mathbf{w}) = \sum_{m=1}^{M} w_m \phi_m(x)$$

- $\mathbf{w} = (w_1, w_2, ..., w_M)$ is the vector of adjustable model parameters ("weights")

- $\widehat{f}(x)$ can be non-linear in x, but is a linear function of $\mathbf{w}$
    - sometimes referred to as “linear in the parameters” (L.I.T.P.)

Slide 4 Basis Functions

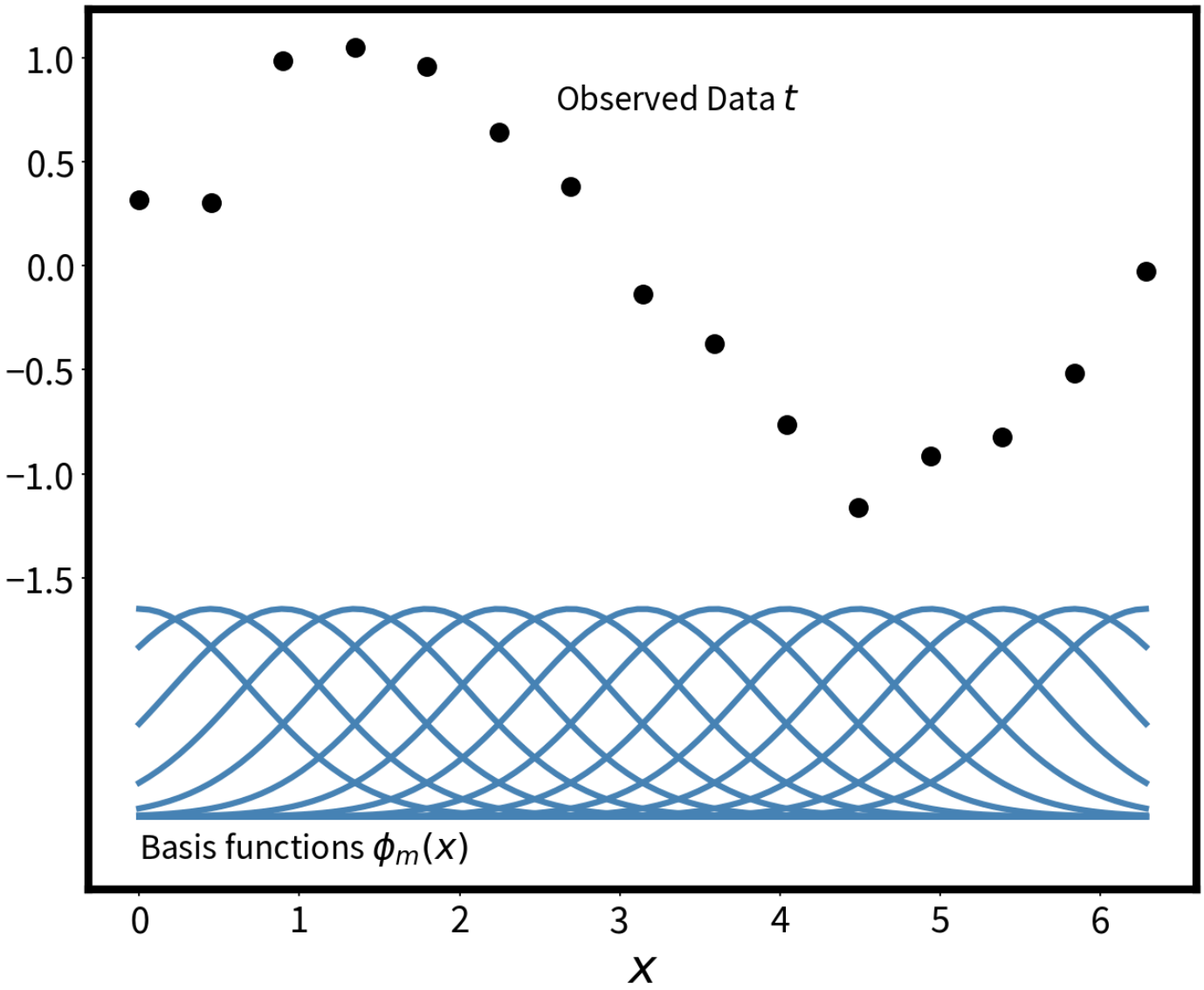
- Many possible choices for basis functions:
    - Linear: $\phi(x) = x$ (makes more sense in higher dimensions)
    - Polynomial: $\phi_m(x) = x^m$
    - Hand-crafted, bespoke, features $\phi_m(x)$
    - General-purpose “Gaussians”, with fixed centres c_m and common radius r:

$$\phi_m(x) = \exp\left\{ -\frac{(x - c_m)^2}{r^2} \right\}$$

      e.g. a radial basis function, RBF, model

- Here we adopt Gaussian radial basis functions, centred on the data, with $c_m = x_m$

Slide 5 Basis Functions Illustrated



Least-Squares Fitting

Slide 6 Model-Fitting: Least-Squares

- Classic method for fitting this model: least-squares
- Minimise the error measure:

$$E_{\mathcal{D}}(\mathbf{w}) = \frac{1}{2}\sum_{n=1}^{N}\left[y_n - \sum_{m=1}^{M} w_m \phi_m(x_n)\right]^2$$

- If we define:
  - Target vector $\mathbf{y} = \left(y_1, ..., y_N \right)^{\mathrm{T}}$
  - Design matrix $\mathbf{\Phi}$, $N \times M$, such that $\mathbf{\Phi}_{nm} = \phi_m \left(x_n \right)$
- Then $E_{\mathcal{D}} \left(\mathbf{w} \right) = \frac{1}{2} \| \mathbf{y} - \mathbf{\Phi w} \|^2$ and the least-squares parameter estimate is:

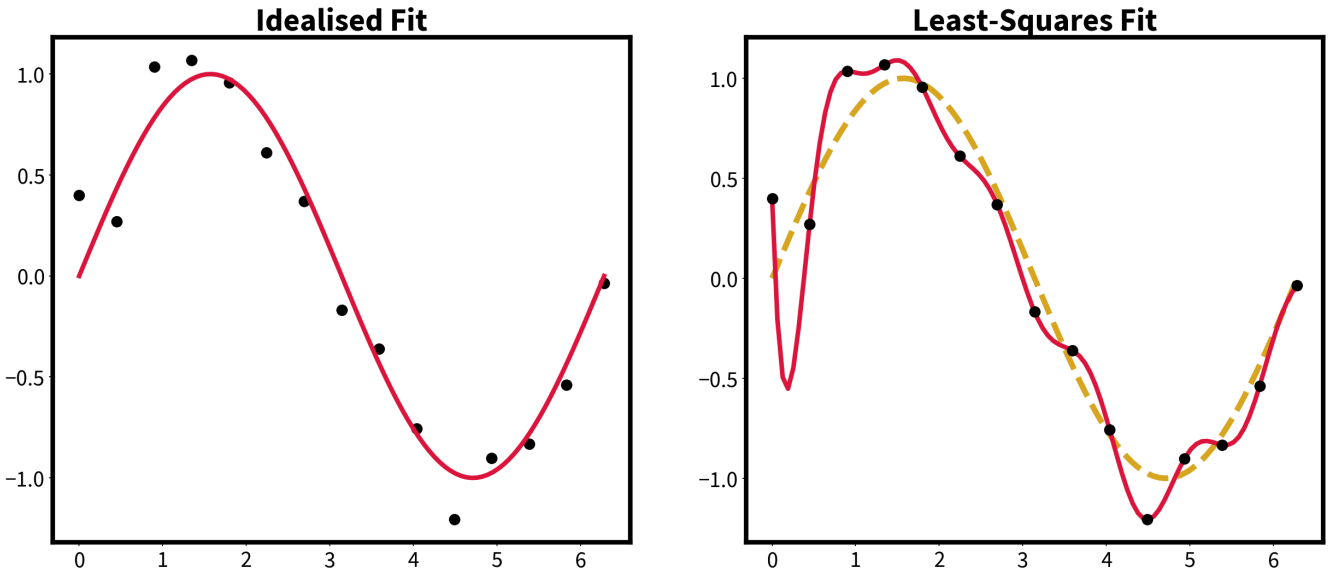
$$\widehat{\mathbf{w}_{LS}} = \left(\mathbf{\Phi}^{\mathrm{T}} \mathbf{\Phi} \right)^{-1} \mathbf{\Phi}^{\mathrm{T}} \mathbf{y}$$

- Since $M = N = 15$ here, this model should be capable of perfectly interpolating all the data points

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Slide 7 Least-Squares Fit

- With $M = 15$ basis functions and only $N = 15$ samples, minimisation of squared-error leads to over-fitting (or ... does it?)



- Which of these is the “better” fit? (There is No Free Lunch!)
- It seems we need some complexity control or regularisation

Penalised Least-Squares

Slide 8 Complexity Control: Regularisation

- We typically think that smoother functions are more realistic, and these generally have smaller weight values $\mathbf{w}$
- Add a weight penalty term to the error function:

$$E_{PLS}(\mathbf{w}) = E_{\mathcal{D}}(\mathbf{w}) + \lambda . E_W(\mathbf{w})$$

- A standard choice of regulariser is the squared-weight penalty:

$$E_W(\mathbf{w}) = \frac{1}{2}\sum_{m=1}^{M} w_m^2$$

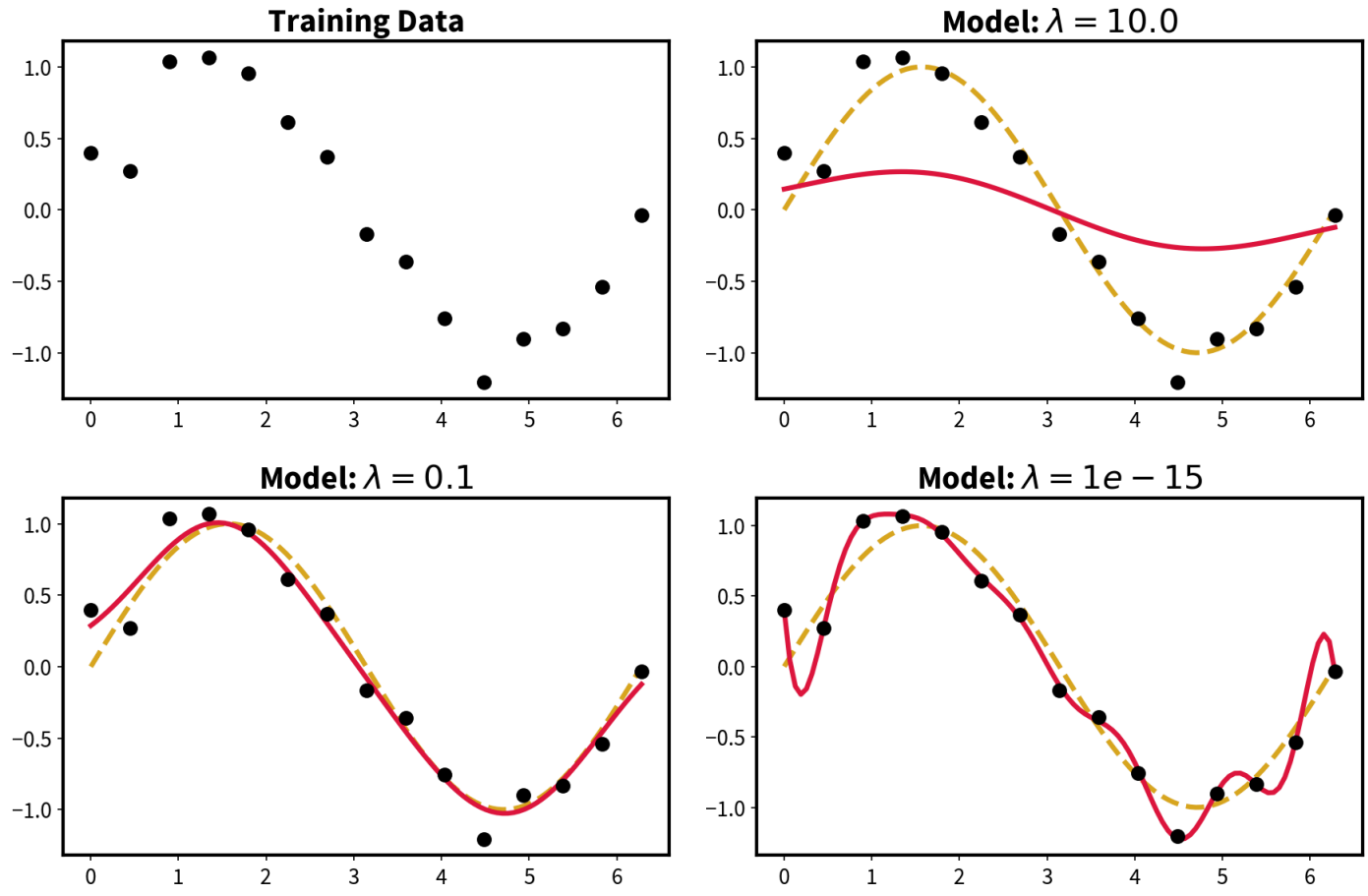
- This gives the penalised least-squares (PLS) estimate:

$$\widehat{\mathbf{w}_{PLS}} = \left(\mathbf{\Phi}^{\mathrm{T}}\mathbf{\Phi} + \lambda\mathbf{I}\right)^{-1}\mathbf{\Phi}^{\mathrm{T}}\mathbf{y}$$

- Also referred to as ridge regression

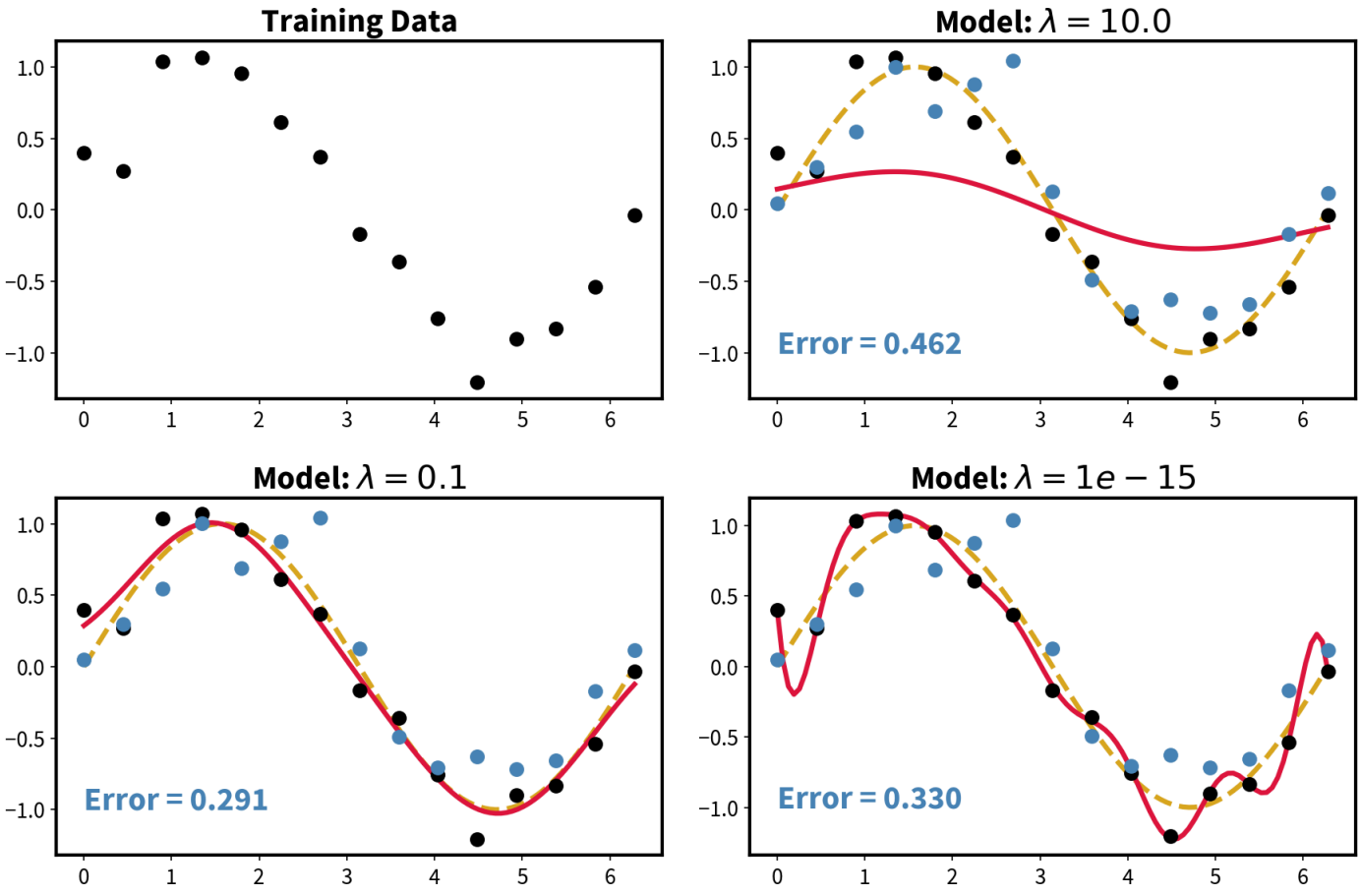
Slide 9 Regularisation: Effect of λ

The hyper-parameter λ balances the trade-off between $E_{\mathcal{D}}$ and E_W:between how well the function fits the data and how smooth it is

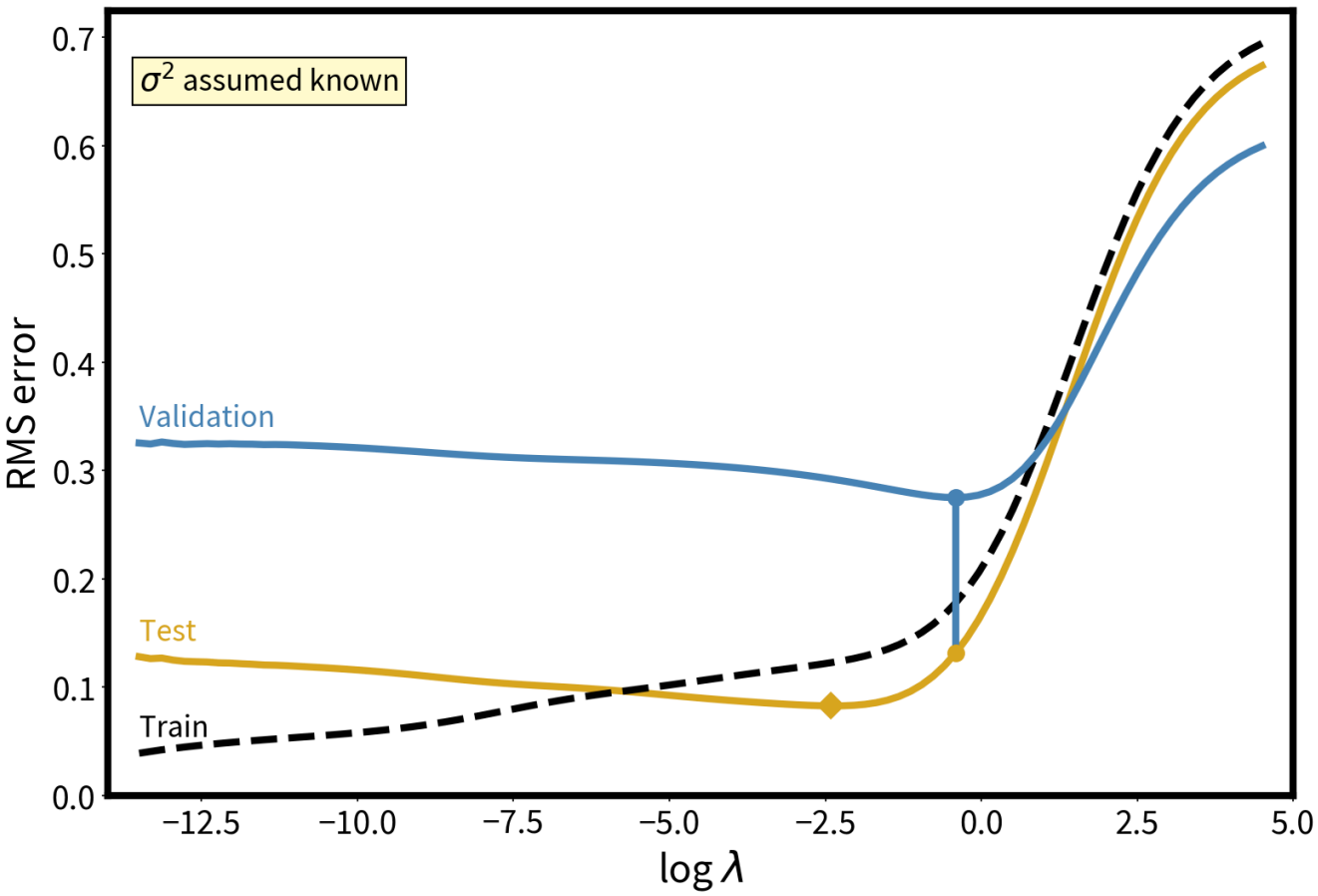


Slide 10 **Regularisation: Validation of λ**

An appropriate value of λ might be selected by assessing the model error on a separate validation set (or hold-out set)


Training Data
Model: $\lambda = 10.0$
Error = 0.462
Model: $\lambda = 0.1$
Error = 0.291
Model: $\lambda = 1e-15$
Error = 0.330

Slide 11 Regularisation: Validation of λ (Over a broad range)



The Probabilistic Framework

Slide 12 A Probabilistic Approach

- Standard (generative) assumption: the data is a noisy realisation of an underlying functional model:

$$y_n = \widehat{f}\left(x_n; \mathbf{w}\right) + \epsilon_n$$

where $\widehat{f}\left(x_n; \mathbf{w}\right)$ is our linear model and ϵ_n the noise (or error)

- *Explicitly* model the noise ϵ_n probabilistically, as $N\left(0, \sigma^2\right)$: a zero-mean Gaussian distribution with variance σ^2

- Since $y_n = \widehat{f}\left(x_n; \mathbf{w}\right) + \epsilon_n$, then $p\left(y_n | \mathbf{w}, \sigma^2\right) = N\left(\widehat{f}\left(x_n; \mathbf{w}\right), \sigma^2\right)$

- Assuming i.i.d. data (as usual), the observation model (or likelihood) is:

$$p\left(\mathbf{y}|\mathbf{w},\sigma^2\right) = \prod_{n=1}^{N} p\left(y_n|\mathbf{w},\sigma^2\right) = \left(2\pi\sigma^2\right)^{-N/2} \prod_{n=1}^{N} \exp\left\{-\frac{\left[y_n - \hat{f}\left(x_n;\mathbf{w}\right)\right]^2}{2\sigma^2}\right\}$$

Slide 13 Maximum Likelihood and Least-Squares

- The maximum-likelihood estimate (M.L.E.) for $\mathbf{w}$ is that value which maximises $p\left(\mathbf{y}|\mathbf{w},\sigma^2\right)$
- This is identical to the least-squares solution $\widehat{\mathbf{w}_{LS}}$ (unregularised)
- To see this, note that:
  - maximising the likelihood is equivalent to minimising the negative logarithm of the likelihood as $\log\left(p\right)$ is monotonic
  - and this is equivalent to minimising squared-error

$$-\log p\left(\mathbf{y}|\mathbf{w},\sigma^2\right) = \frac{N}{2}\log\left(2\pi\sigma^2\right) + \frac{1}{2\sigma^2}\sum_{n=1}^{N}\left[y_n - \widehat{f}\left(x_n;\mathbf{w}\right)\right]^2$$

- The quantity $-\log p\left(\mathbf{y}|\mathbf{w},\sigma^2\right)$ is the negative log-likelihood

Bayesian Regression Framework

Slide 14 The Bayesian Framework: Incorporating Priors

- Instead of the regularisation weight penalty $E_W\left(\mathbf{w}\right)$, we now control the model's complexity via a prior distribution which expresses our degree of belief, or our preference, over values that $\mathbf{w}$ might take
- A convenient (conjugate) choice is:

$$p\left(\mathbf{w}|\alpha\right) = \prod_{m=1}^{M}\left(\frac{\alpha}{2\pi}\right)^{1/2}\exp\left\{-\frac{\alpha}{2}w_m^2\right\}$$

- This is a zero-mean Gaussian prior, independent for each weight, with common inverse variance (precision) hyper-parameter α

Key feature of the Bayesian framework Rather than penalising larger weights, we express a preference for smoother models by declaring smaller weights to be *a priori* more probable

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Slide 15 **Bayesian Inference**

- Given the likelihood and the prior, rather than computing a single *point estimate* $\widehat{\mathbf{w}}_{PLS}$ for the weights, we compute the posterior distribution via Bayes' rule:

$$p\left(\mathbf{w}|\mathbf{y},\alpha,\sigma^2\right) = \frac{\text{likelihood}\times\text{prior}}{\text{normalising factor}} = \frac{p\left(\mathbf{y}|\mathbf{w},\sigma^2\right)\,p\left(\mathbf{w}|\alpha\right)}{p\left(\mathbf{y}|\alpha,\sigma^2\right)}$$

- Here, since the prior is conjugate, the posterior is conveniently also Gaussian: $p\left(\mathbf{w}|\mathbf{y},\alpha,\sigma^2\right) = N\left(\boldsymbol{\mu},\boldsymbol{\Sigma}\right)$ with

$$\begin{aligned}\boldsymbol{\mu} &= \left(\boldsymbol{\Phi}^{\mathrm{T}}\boldsymbol{\Phi} + \sigma^2\alpha\mathbf{I}\right)^{-1}\boldsymbol{\Phi}^{\mathrm{T}}\mathbf{y}\\ \boldsymbol{\Sigma} &= \sigma^2\left(\boldsymbol{\Phi}^{\mathrm{T}}\boldsymbol{\Phi} + \sigma^2\alpha\mathbf{I}\right)^{-1}\end{aligned}$$

"Bayesian" MAP Estimation

Slide 16 **MAP Estimation — a "Bayesian" Short-cut**

- The maximum *a posteriori* (MAP) estimate for $\mathbf{w}$ is the single most probable value under the posterior $p\left(\mathbf{w}|\mathbf{y},\alpha,\sigma^2\right)$

- Since the denominator in Bayes' rule earlier is independent of $\mathbf{w}$, this is equivalent to minimising:

$$E_{MAP}\left(\mathbf{w}\right) = -\log p\left(\mathbf{y}|\mathbf{w},\sigma^2\right) - \log p\left(\mathbf{w}|\alpha\right)$$

- Retaining only terms which depend on $\mathbf{w}$:

$$E_{MAP}\left(\mathbf{w}\right) = \frac{1}{2\sigma^2}\sum_{n=1}^{N}\left\{y_n - \widehat{f}\left(x_n;\mathbf{w}\right)\right\}^2 + \frac{\alpha}{2}\sum_{m=1}^{M}w_m^2$$

- The MAP estimate is therefore identical to the PLS estimate with $\lambda = \sigma^2\alpha$. As a result: $\boldsymbol{\mu} = \widehat{\mathbf{w}_{PLS}}$

Step-by-Step Example of Bayesian Learning

Slide 17 Example: Bayesian "Learning"

- Let's look at how the posterior $p\left(\mathbf{w}|\mathbf{y}, \alpha, \sigma^2\right)$ evolves as we observe data y_n
  - for demonstration purposes, α and σ^2 are set to some fixed values here
- Note that we can compute the posterior incrementally if the data are assumed independent (given $\mathbf{w}$). *e.g.* for $\mathbf{y} = \left(y_1, y_2, y_3\right)$:

$$\begin{aligned} p\left(\mathbf{w}|y_1, y_2, y_3\right) &\propto p\left(y_1, y_2, y_3|\mathbf{w}\right) p\left(\mathbf{w}\right) \\ &\propto p\left(y_2, y_3|\mathbf{w}\right) p\left(y_1|\mathbf{w}\right) p\left(\mathbf{w}\right) \\ &\propto \text{Likelihood of } \left(y_2, y_3\right) \times \text{posterior given } y_1 \end{aligned}$$

- So, more generally, we can treat the posterior having observed $\left(y_1, ..., y_K\right)$ as the "prior" for the remaining data $\left(y_{K+1}, ..., y_N\right)$

Slide 18 Sequential Bayesian Learning

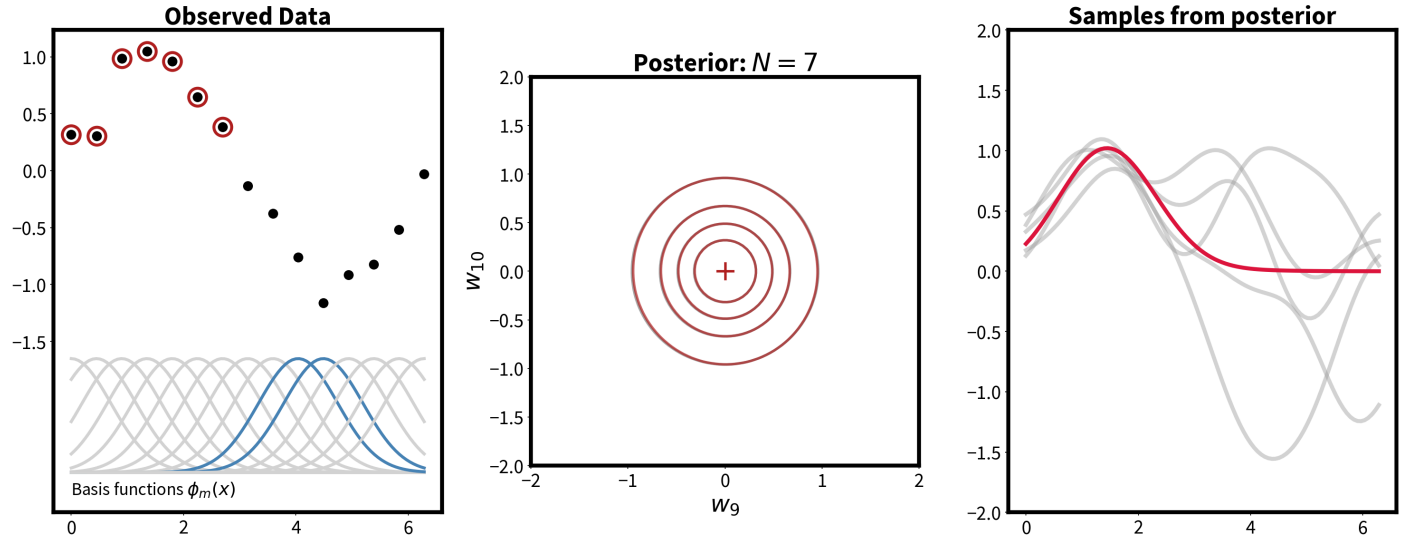
The prior distribution, before observing data.

We focus on weights 9 and 10 (basis functions highlighted in blue).



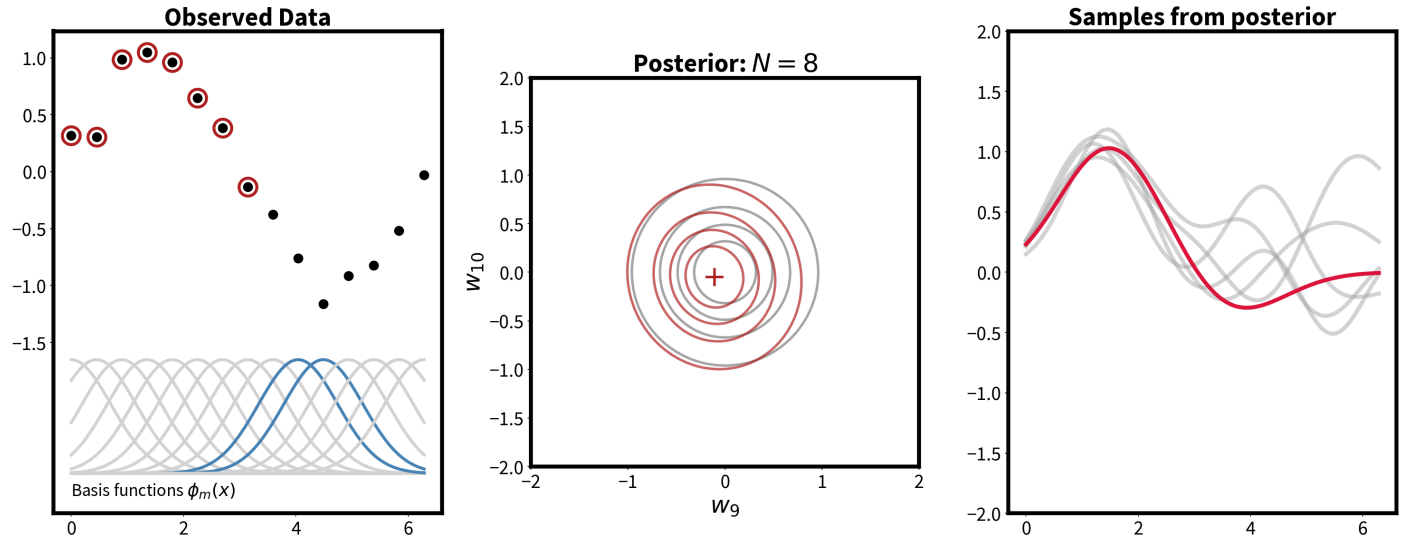
Slide 19 Sequential Bayesian Learning

The posterior distribution, after observing 7 data points ...



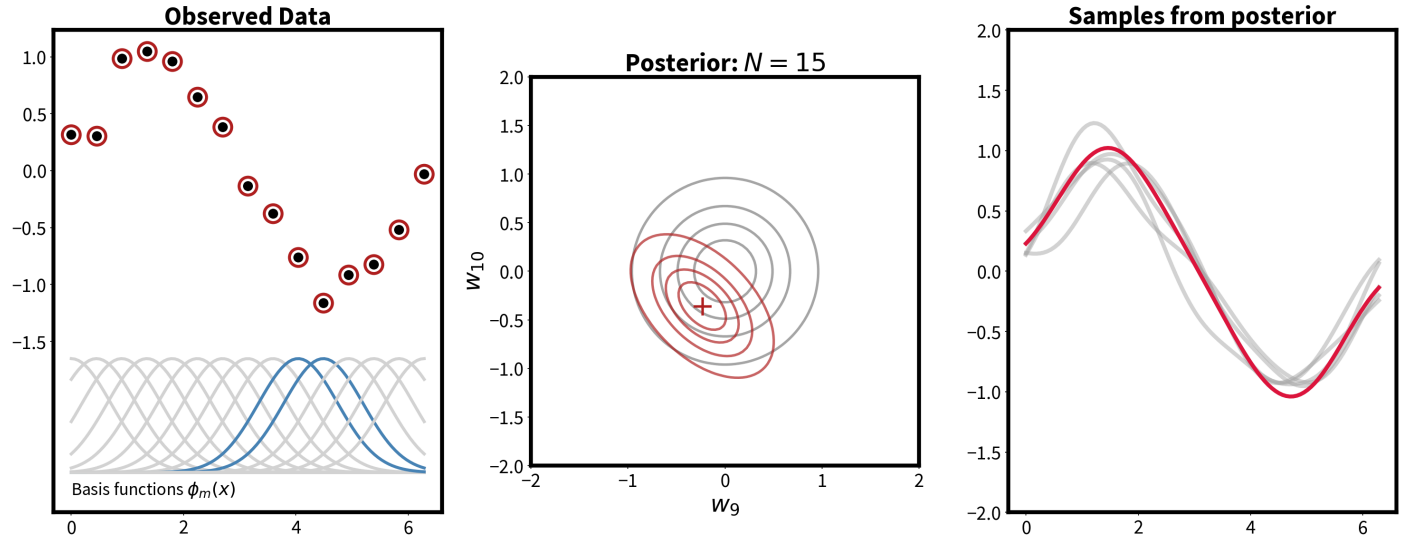
Slide 20 Sequential Bayesian Learning

The posterior distribution, after observing 8 data points ...



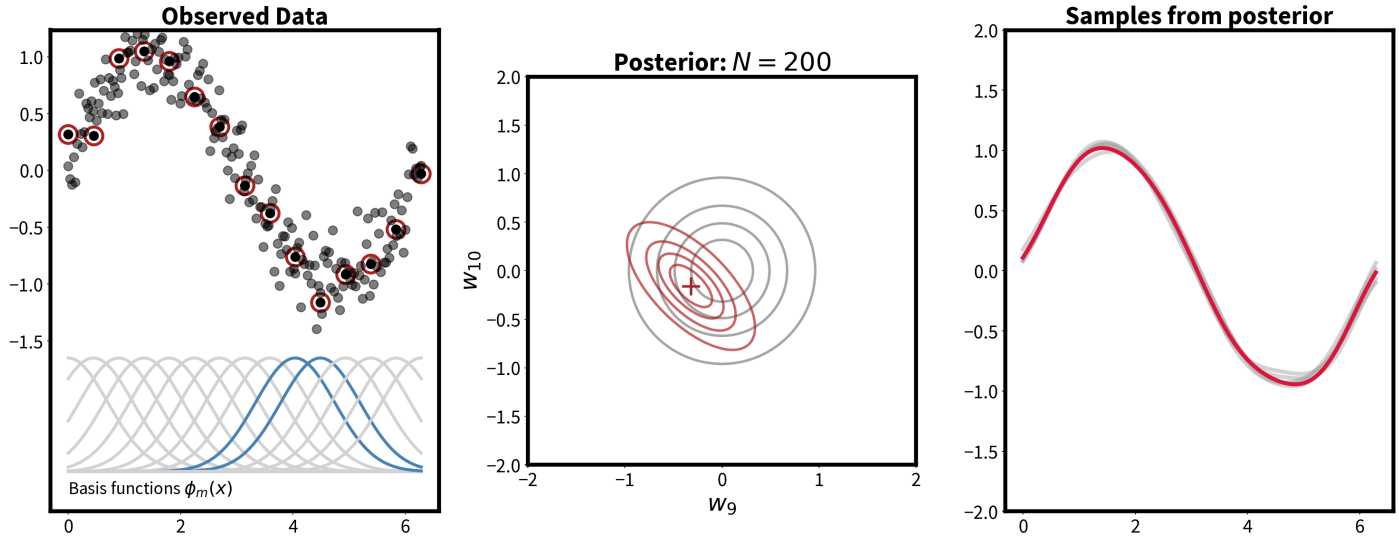
Slide 21 Sequential Bayesian Learning

The posterior distribution, after observing 15 data points ...



Slide 22 Sequential Bayesian Learning

The posterior distribution, after observing 200 data points ...



Summary

Slide 23 Summary

- It may appear that we have simply repeated least-squares / maximum likelihood methods within a different (more complex!) framework
- So … what was the point?
- In the next lecture, we will build on this and see the real practical benefits of the Bayesian approach:
  - taking account of uncertainty in $\mathbf{w}$
  - finding a good setting for α *without needing a validation set*
  - ... while simultaneously estimating σ^2
- Later, we will look at the choice of the model itself
  - for example: setting the “fixed” basis function width r